

Tree and Grammar Induction

★ note how we split each tree of height k into smaller subtrees of height $\leq k-1$

$$\sum_{i=0}^k i \rightarrow \sum_{i=0}^{k-1} i + k \quad \text{split for summation}$$

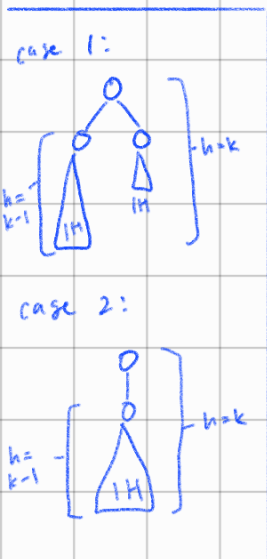
ex. Let T be binary tree height k , n nodes. Prove by induction that $n \leq 2^{k+1} - 1$.

Proof: Prove by induction on height h .

Base case - $h=0$ is tree w/ single node, $n=1$, $2^{0+1}-1=1$, $1 \leq 1$ so base case holds.

Inductive hypothesis - Suppose binary trees w/ height h , n nodes have $n \leq 2^{h+1} - 1$ for $h=0, 1, \dots, k-1$.

Inductive Step - We want to show binary tree height k has $n \leq 2^{k+1} - 1$. Suppose we have binary tree of height k . We have two cases.



Case 1: Root node has two children, each of which are the roots of a binary tree of height $\leq k-1$. By IH, the left subtree has $n_L \leq 2^k - 1$ and right subtree has $n_R \leq 2^k - 1$ where n_L and n_R are # nodes in L/R subtrees. Then, our tree has $1 + n_L + n_R = n$ many nodes. So $n \leq 2^k - 1 + 2^k - 1 + 1 = 2 \cdot 2^k - 1 = 2^{k+1} - 1$, $n \leq 2^{k+1} - 1$.

Case 2: Root node has only one child, which is the root of binary tree of height $= k-1$. By IH, $n' \leq 2^k - 1$ where n' is # of nodes in the subtree. Then our tree has $n = n' + 1$, so $n \leq 2^k - 1 + 1 = 2^k \leq 2^{k+1} - 1$, $n \leq 2^{k+1} - 1$ since $k > 0$.

Since in both cases, $n \leq 2^{k+1} - 1$, we have proven our claim.

Grammar Tree Induction

- claim about tree $\rightarrow$ induction on height
- claim about string (terminal seq.) $\rightarrow$ induction on string length

ex. Grammar G w/ start S and terminals a, b w/ rules: $S \rightarrow ab | SS | aSb$. Prove all trees generated by G have same # of a and b nodes

eg. Proof: Prove by induction on height of tree h .



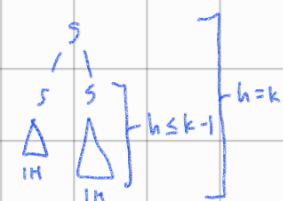
$h=0$ invalid
base case by rules

Base case - Binary tree w/ $h=1$, has 1 a and 1 b so base case holds.

Inductive hypothesis - All binary trees of height h generated by G have equal # a 's and b 's for $h=1, 2, \dots, k-1$

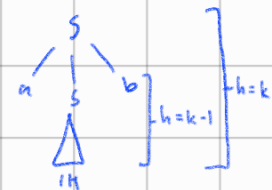
Inductive step - want to show tree of height k generated by G has equal # a 's and b 's. Suppose binary tree height k produced by G . We have two cases.

case 1:



Case 1: Root node S has two children according to rule 2, call them S_1 and S_2 . They are root nodes of height $\leq k-1$ produced by G , so by IH, tree rooted in S_1 has m -many a 's and b 's where $m, j \in \mathbb{Z}$ and tree rooted in S_2 has j -many a 's and b 's. Our tree then has $m+j$ many a 's and b 's.

case 2:



Case 2: Root node S has 3 children according to rule 3 of grammar - a, S_1, b . Subtree rooted at S_1 has height $= k-1$ so we can apply IH. Say subtree has m many a 's and b 's where $m \in \mathbb{Z}$, our tree has $m+1$ many a 's and b 's.

In both cases, number of a 's and b 's are equal, so we have proven our claim.

Discussion Problems - 13.3b, 13.2a

(b) A parity tree is a full binary tree with each node colored orange or blue, such that:

- (1) If v is a leaf node, then v is colored orange.
- (2) If v has two children of the same color, then v is colored blue.
- (3) If v has two children of different colors, then v is colored orange.

Prove by induction that every parity tree has the *parity property*: if the root is colored orange, then it has an odd number of leaves; and if the root is colored blue, then it has an even number of leaves.

b) Proof: We will prove by induction on h , the height of the ^{full binary with n leaf nodes} tree T , that every parity tree has the parity property. For our base case, consider a binary tree with $h=0$, which is a single root node. Because the single node is a leaf node, by rule 1, that single node is orange. The base case then holds because the number of nodes is odd (1) and the root is orange. For our inductive step, suppose that the parity property holds for all trees of height $h=0, 1, \dots, k-1$. For our inductive step, we're trying to prove that the parity property holds for a tree of height $h=k$. We have three cases. Let T_L and T_R be the left and right subtrees where the roots of the subtrees are the left and right children of the root node. Let n_L and n_R be the number of leaves of T_L and T_R . By IH, we know parity property holds for T_L and T_R because their height $\leq k-1$.

Case 1: n_L and n_R are both even. By the parity property, the roots of T_L and T_R must both be blue. Because the roots of T_L and T_R are the children of the root of T and they're both blue, by rule 2, the root of T is blue. $n_L + n_R = n$, and n is even because n_L and n_R are even and the sum of two even numbers is even. The tree has the parity property for case 1 because the root is blue and the number of leaves is even.

Case 2: n_L and n_R are both odd. By the parity property, the roots of T_L and T_R must both be orange. Because the roots of T_L and T_R are the children of the root of T and they're both orange, by rule 2, the root of T is blue. $n_L + n_R = n$, and n is even because n_L and n_R are odd and the sum of two odd numbers is even. The tree has the parity property for case 2 because the root is blue and the number of leaves is even.

Case 3: n_L is odd and n_R is even. By the parity property, the root of T_R must be blue and the root of T_L must be orange. Because the roots of T_L and T_R are the children of the root of T and they're different colors, by rule 3, the

root of T is orange. $n_L + n_R = n$, and n is odd because n_L is odd and n_R is even, and the sum of an odd and even number is odd. The tree has the parity property for case 3 because the root is orange and the number of leaf nodes is odd.

Case 4: n_L is even and n_R is odd. By the parity property, the root of T_L must be blue and the root of T_R must be orange. Because the roots of T_L and T_R are the children of the root of T and they're different colors, by rule 3, the root of T is orange. $n_L + n_R = n$, and n is odd because n_R is odd and n_L is even, and the sum of an odd and even number is odd. The tree has the parity property for case 4 because the root is orange and the number of leaf nodes is odd.

In all cases, T has the parity property, so claim is proven.

- (a) Define a grammar G_1 by $S \rightarrow aSbS \mid SaS \mid ab \mid a$. where S is the only start symbol and the terminal symbols are a and b . Prove that a tree generated by G_1 has at least as many nodes labeled a as nodes labeled b .



Proof: We will prove by induction on height h that a tree generated by G_1 has at least as many a nodes as b nodes. For our base case, we have a tree with $h=1$ and either a single a node connected to the root S node or an a and b node connected to the root S node (rules 3 and 4). In either case, the number of a nodes is greater than or equal to the number of b nodes, so the base case holds. For our inductive hypothesis, suppose a binary tree generated by G_1 has at least as many a nodes as b nodes for height $h=1, 2, 3, \dots, k-1$. For our inductive step, we're trying to show that the tree T with $h=k$ has at least as many a nodes as b nodes. Let T_1 and T_2 be the subtrees formed by rule 1 and 2, rooted in the children of T . The height of T_1 and T_2 is $\leq k-1$, so we can apply the inductive hypothesis to T_1 and T_2 to say that in both subtrees, there are

at least as many a's as b's. Now we can look at two possible cases.

Case 1: T follows rule 1 and the children of the root are a, S, b, S . We already know that for both subtrees T_1 and T_2 rooted in the S 's, there are at least as many a nodes as b nodes. If we add 1 a and 1 b node, there are still at least as many a nodes as b nodes.

Case 2: T follows rule 2 and the children of the root are S, a, S . We already know that for both subtrees T_1 and T_2 rooted in the S 's, there are at least as many a nodes as b nodes. If we add 1 a node, there are more a nodes than b nodes, which still means there are at least as many a nodes as b nodes.

In either case, there are at least as many a nodes as b nodes so the claim is proven.